EduPro Online Learning Dashboard: A Data Analytics and Visualization Approach Using Python and Streamlit

Author

Pramit Nandi

Internship Project

Abstract

The rapid growth of online learning platforms has generated vast amounts of educational data that can be analyzed to improve decision-making and user experience. This project presents the design and development of an interactive dashboard for the EduPro Online Learning Platform using Python and Streamlit. The dashboard integrates data from multiple sources, including users, teachers, courses, and transactions, to provide meaningful insights through interactive visualizations. Data preprocessing, cleaning, merging, and visualization techniques were applied using Pandas, Matplotlib, Seaborn, and Plotly. The dashboard enables users to monitor key performance indicators (KPIs), analyze revenue trends, evaluate teacher performance, and understand user behavior.

Keywords: Data Analytics, Streamlit, Python, Dashboard, Pandas, Plotly, Online Learning

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1. Introduction

Online learning platforms have become an essential part of modern education. They produce large volumes of data related to student enrollments, course performance, instructor ratings, and financial transactions. Analyzing this data helps organizations make informed business decisions.

The objective of this project is to build an interactive dashboard that allows users to analyze educational data efficiently using modern data visualization techniques.

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2. Objectives

The primary objectives of this project are:

- Analyze educational platform data.
- Perform data cleaning and preprocessing.
- Merge multiple datasets into a single master dataset.
- Generate meaningful business insights.

- Develop an interactive dashboard using Streamlit.
- Visualize trends using interactive charts.

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3. Dataset Description

The project uses four datasets stored in an Excel workbook.

Users

Contains user information such as:

- User ID
- User Name
- Age
- Gender
- Email

Teachers

Contains teacher information including:

- Teacher ID
- Teacher Name
- Expertise
- Experience
- Teacher Rating

Courses

Contains course information:

- Course Name
- Category
- Level
- Price
- Rating
- Duration

Transactions

Contains purchase records:

- Transaction ID
- User ID
- Course ID
- Teacher ID
- Payment Method
- Amount
- Transaction Date

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4. Methodology

The project was completed through the following stages:

Step 1

Data Collection

The Excel dataset was imported using the Pandas library.

Step 2

Data Cleaning

The dataset was cleaned by:

- Removing duplicate records
- Checking missing values
- Converting date formats
- Renaming duplicate columns after merging

Step 3

Data Integration

Multiple datasets were merged using:

- UserID
- CourseID
- TeacherID

Step 4

Exploratory Data Analysis (EDA)

Various analyses were performed:

- Gender Distribution
- Age Distribution
- Revenue Analysis
- Payment Method Analysis
- Course Category Analysis
- Monthly Revenue Trend
- Teacher Performance

Step 5

Dashboard Development

The dashboard was developed using Streamlit with interactive charts created using Plotly.

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5. Tools and Technologies

Technology	Purpose
Python	Programming Language
Pandas	Data Processing
NumPy	Numerical Computation
Streamlit	Dashboard Development
Plotly	Interactive Visualization
Matplotlib	Data Visualization
Seaborn	Statistical Visualization

Technology	Purpose
OpenPyXL	Excel File Reading
PyCharm	IDE

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6. Dashboard Features

The dashboard includes:

- Dashboard Summary Cards
- Total Users
- Total Teachers
- Total Courses
- Total Revenue
- Sidebar Filters
- Revenue by Course Category
- Revenue by User Gender
- Payment Method Distribution
- Monthly Revenue Trend
- Top 10 Teachers
- Top 10 Courses
- Download Dataset Button
- Interactive Charts

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7. Results

The dashboard successfully provides business insights including:

- Overall revenue generated
- Popular course categories
- User demographic analysis

- Teacher performance evaluation
- Payment method preferences
- Monthly revenue growth
- Top-performing courses

These insights can help management improve marketing strategies, course offerings, and user engagement.

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8. Advantages

- Interactive dashboard
- Easy-to-use interface
- Real-time filtering
- Better business insights
- Faster decision-making
- Scalable architecture
- Professional visualization

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9. Limitations

- Uses a static dataset.
- No real-time database integration.
- Authentication is not implemented.
- Predictive analytics is not included.

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10. Future Scope

Future improvements include:

- Database integration using MySQL or PostgreSQL
- User authentication system
- Machine Learning-based course recommendation
- Sales prediction using forecasting models

- Cloud deployment
- Mobile-friendly dashboard
- AI-powered analytics

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11. Conclusion

This project demonstrates how Python and Streamlit can be used to transform raw educational data into meaningful visual insights. By integrating multiple datasets and presenting them through an interactive dashboard, users can analyze business performance efficiently. The project provides practical experience in data cleaning, analysis, visualization, and dashboard development, making it a valuable learning experience for data analytics and business intelligence applications.

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